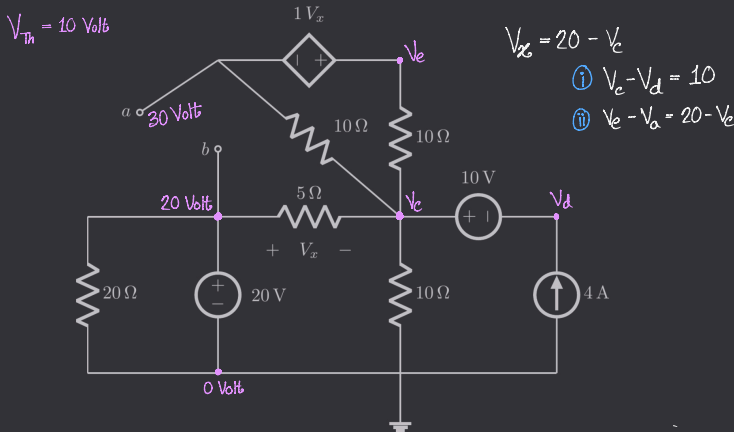


Question 1.

Obtain the Thevenin or Norton equivalent circuit with respect to terminals a-b and find the maximum power that can be supplied to any load connected to a-b.



iii KCL @ node - c: $4 = \frac{V_c - V_a}{10} + \frac{V_c - V_e}{10} + \frac{V_c - 20}{5} + \frac{V_c - 0}{10}$

$$40 = 5V_c - V_a - V_e - 40$$

$$80 = 5V_c - V_a - V_e$$

iv KCL @ supernode (a-e): $\frac{V_a - V_c}{10} + \frac{V_e - V_c}{10} = 0$

$$V_a + V_e - 2V_c = 0$$

$$2V_c - V_a - V_e = 0$$

$$V_c = \frac{80}{3}$$

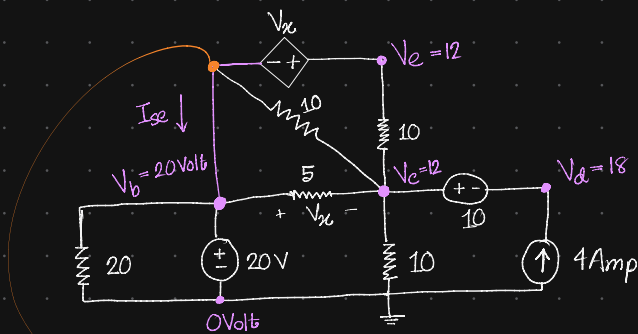
$$V_a = 30$$

$$V_e = \frac{70}{3}$$

$$V_d = \frac{50}{3} \text{ Volt}$$

$$V_{Th} = 10$$

Calculating I_{sc}



KCL here: $0 = \frac{V_b - V_e}{10} + \frac{V_e - V_c}{10} + I_{sc}$

$$0 = \frac{20 - \frac{70}{3}}{10} + \frac{\frac{50}{3} - \frac{70}{3}}{10} + I_{sc}$$

$$0 = -1 + I_{sc}$$

$$I_{sc} = -1 \text{ Amp}$$

$$R_{Th} = \frac{V_{Th}}{I_{sc}} = \frac{10}{-1} = 10$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{10^2}{4 \times 10} = 2.5 \text{ watt}$$

$$i_x = \frac{V_g - V_e}{1} = -V_e$$

KCL @ (c-d) Supernode :

$$\frac{V_c - 3}{3} + \frac{V_c - 0}{1} + \frac{V_d - 0}{1} + \frac{V_d - V_a}{3} = 0$$

KEL @ node a

$$V_c - 3 + 3V_c - 9V_c - 3V_c - V_a = 0$$

$$-8V_c - V_a = 3$$

$$\delta V_c + V_a = -3$$

$$-V_a + 8V_c = -2$$

$$1 = \frac{V_a - V_d}{3} + \frac{V_a - 3}{24}$$

$$24 = 8V_a + 24V_c + V_a - 3$$

$$9V_a + 24V_c = 27$$

$$3V_a + 8V_c = 9$$

$$V_a = 6 \text{ Volt}$$

$$V_e = -\frac{9}{8} \text{ Volt}$$

$$V_d = \frac{27}{8} \text{ Volt}$$

$$V_{Th} = 3$$

Shorting $(a-b)$:

- $V_a = V_b = 3 \text{ Volt}$

- 24 ohm gone/shorted

- KCL @ (c-d) Supernode :

$$V_a + 8V_c = -3$$

$$3 + 8V_c = -3$$

$$8V_f = -6$$

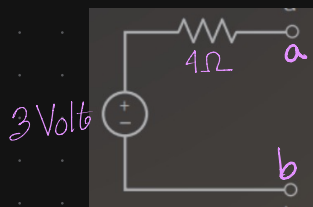
$$V_2 = -6/8$$

- $V_d = -2V_c = \frac{18}{8} = \frac{9}{4}$

$$I_{se} = 1 + \frac{V_d - V_a}{2} = 1 + \frac{9/4 - 3}{2} = 3/4$$

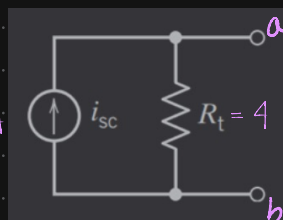
$$R_{Th} = \frac{V_{Th}}{I_{SC}} = \frac{2}{3/4} = 4$$

Ques 2.a)



2. b)

$\frac{3}{4} A$



Ques-3

$$\begin{matrix} V & I \\ 2 & 3 \\ -6 & -1 \end{matrix} \quad \frac{\Delta I}{\Delta V} = \frac{4}{8}$$

$$I = \frac{1}{2}V + 2 \Rightarrow V = 2I - 4$$

$$R_{eq} = 2$$

$$V_{Th} = -4 \text{ Volt}$$

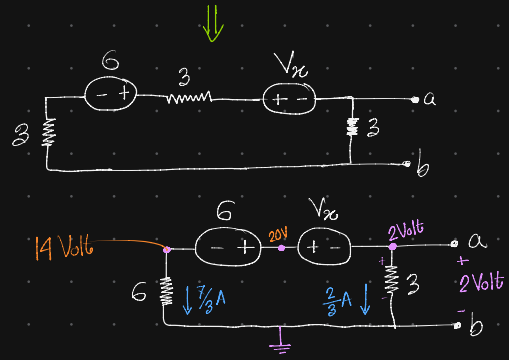
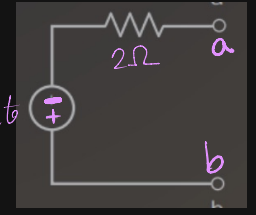
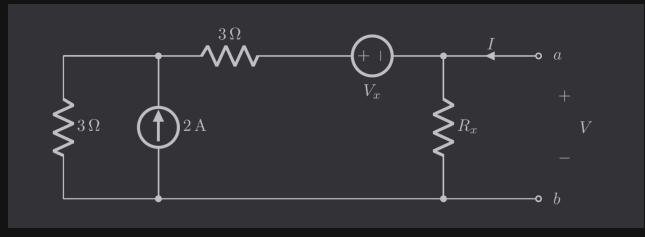
$$(3+3) \parallel R_x = R_{eq} = 2$$

$$\frac{6R_x}{6+R_x} = 2$$

$$6R_x = 12 + 2R_x$$

$$4R_x = 12$$

$$R_x = 3 \Omega$$



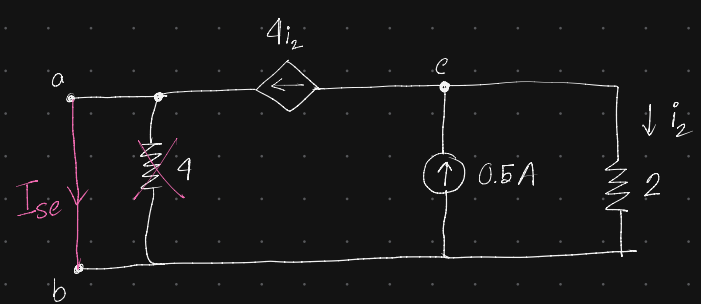
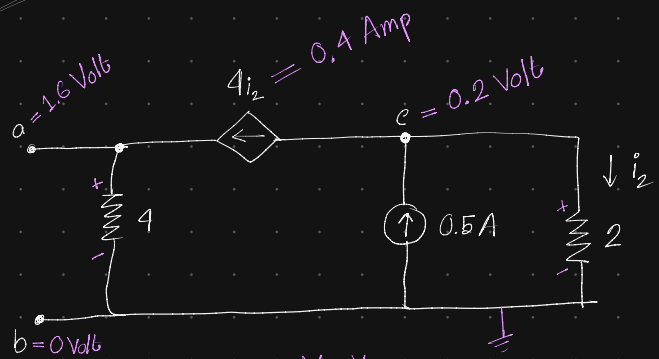
if $V_{ab} = 2$ then $I = 3$

$$i_{3\Omega} = \frac{2}{3}$$

$$i_{6\Omega} = I - \frac{2}{3} = \frac{7}{3}$$

• $\Rightarrow$ implies • $\Rightarrow$ implies • $\Rightarrow V_x = 18$

Ques 4



$$4i_2 = \frac{V_a - V_b}{4}$$

KCL @ c

$$4i_2 + i_2 = 0.5 \text{ Amp}$$

$$i_2 = 0.1 \text{ Amp}$$

KCL @ c

$$4i_2 + i_2 = 0.5 \text{ Amp}$$

$$i_2 = 0.1 \text{ Amp}$$

$$I_{se} = 0.4 \text{ Amp}$$

$$V = IR$$

$$V_{ab} = 0.4 \times 4$$

$$V_{ab} = V_{Th} = 1.6 = V_b$$

$$R_{eq} = \frac{1.6}{0.4} = 4 \Omega = R_b$$

$$V = 4I + 1.6$$

$$I = \frac{1}{4}V - 0.4$$

No matter which kind of Load we connect
Dependent Source's current did not change
by our analysis.

